CHAPTER 2 - LITERATURE REVIEW

2.1 Introduction

This chapter reviews research, existing software and relevant technologies related to Work-Integrated Learning (WIL) documentation and progress monitoring. The review considers how digital systems support students, workplace mentors, coordinators and lecturers, and identifies features that can inform the design of TrackWIL.

The chapter compares three existing systems: InPlace, Sonia Online and PebblePad/Mahara-style ePortfolio tools, against the project's needs. It also considers cloud delivery, database-backed web applications, role-based access and secure document upload. The findings are used to define a practical research gap and guide TrackWIL's proposed technical direction

2.2 Existing Research

Work-Integrated Learning connects academic learning with practical workplace experience. Technology can support WIL in different ways: it may deliver learning activities, or it may provide administrative support for placements, communication, evidence collection and assessment. Schuster and Glavas (2017) explain that electronic WIL covers a range of technology uses rather than one single model. For TrackWIL, the administrative role is especially relevant because the project focuses on daily activity records, evidence submission, mentor verification and progress monitoring.

Effective placement supervision depends on communication and coordination between students and supervisors. Hains-Wesson et al. (2026), in a systematic literature review of placement supervision, examine student and supervisor perspectives and identify patterns and gaps in the literature. This supports the need to consider both sides of the submission and feedback process rather than designing a system only for students or academic staff.

Industry systems show that placement management commonly involves centralized records, user portals, communication, compliance and reporting. InPlace describes tools for central placement visibility, student portals, time sheets, communications, compliance and analytics (InPlace Software, n.d.). Sonia Online describes itself as a placement management system containing placement information and paperwork for students and host organizations (Sonia Online, n.d.). These examples show that digital coordination is an established approach, although their exact configuration and available functions depend on the institution using them.

Portfolio platforms also provide relevant ideas for collecting and reviewing evidence. Mahara allows learners to upload evidence, reflect on experiences, share portfolios with selected people and receive feedback (Mahara, n.d.). PebblePad's ATLAS assessment area supports assessors in accessing and reviewing learners' work (PebblePad, 2026). These features are relevant to TrackWIL's evidence and

mentor-review workflow, but a portfolio or assessment platform is not necessarily designed around a specific institution's daily WIL forms and status-tracking requirements.

Overall, the literature and industry examples support a centralized digital workflow, clear communication, evidence management and timely feedback. However, TrackWIL must still be evaluated against its specific project requirements, including daily submissions, mentor approval or rejection, feedback, role-specific dashboards and a visible list of outstanding work. Product websites describe vendor-provided capabilities; they should not be treated as independent proof that a system will improve outcomes in every setting.

2.3 Existing or Similar Systems

Three relevant systems were reviewed: InPlace, Sonia Online and Mahara/PebblePad. The comparison below is based on publicly available product information, not on hands-on testing. Features may vary by institution, subscription or local configuration.

System 1: InPlace: InPlace is designed for higher-education placement management. Its stated features include centralized placement management, student portals, timesheets and schedules, communications, compliance management, integrations and reporting (InPlace Software, n.d.). Its strength is broad administration and visibility across placements. A limitation for this comparison is that its public description emphasizes placement allocation, compliance and programme-level management; it does not establish that every TrackWIL-specific daily logbook or mentor-signing workflow is available in a particular deployment. Lesson for TrackWIL: centralize records and provide role-appropriate views, but prioritize the daily documentation workflow required by the project.

System 2: Sonia Online: Sonia Online is presented as a student placement management system for students and host organizations, with placement information and paperwork available through the system (Sonia Online, n.d.). Its strength is that it directly targets placement coordination and provides access for relevant placement participants. Its limitations cannot be fully assessed from the public landing page alone, and access or features may differ between institutions. Lesson for TrackWIL: make student and mentor access straightforward and keep placement paperwork in one organized location.

System 3: Mahara / PebblePad: Mahara supports evidence-based portfolios, reflection, sharing and feedback (Mahara, n.d.). PebblePad's ATLAS area supports access to learner submissions and assessment (PebblePad, 2026). Their strengths are evidence collection, sharing and review. A limitation relative to TrackWIL is that portfolio and assessment workflows may not, by themselves, provide the project's complete set of daily WIL status controls, outstanding-document monitoring and separate coordinator and lecturer dashboards. Lesson for TrackWIL: connect each uploaded evidence item to a dated activity, submission status, mentor feedback and progress overview.

Comparative summary: InPlace is strongest as a broad placement-management platform; Sonia Online is a directly relevant placement-information and paperwork system; Mahara/PebblePad provide useful portfolio and assessment patterns. TrackWIL draws on these ideas but will focus its scope on

daily WIL activity records, supporting evidence, mentor verification, feedback and monitoring by coordinators and lecturers. This is a project-specific design rationale, not a claim that the existing products lack all such features.

2.4 Technology Review

Web-based application: A browser-based application is suitable because students, mentors, coordinators and lecturers may access the system from different locations and devices. A shared application can present role-specific dashboards while keeping records in a central service. Its disadvantages include dependence on network availability, the need to support different screen sizes and the need for secure deployment and maintenance.

Database: A relational database is appropriate for structured relationships between users, roles, students, assigned mentors, daily activity entries, uploaded evidence, review comments and submission statuses. Using linked tables and primary/foreign keys can help preserve consistency and make it possible to report on missing submissions and verification progress. The final database technology should be selected to fit the group's development environment and deployment requirements; Chapter 1 does not specify a final database product.

Authentication and role-based access: TrackWIL handles student records and workplace evidence, so users should authenticate before accessing protected information. Authorization should restrict users to the records and actions needed for their roles, for example, a mentor should access assigned students' submissions, while coordinators and lecturers should have monitoring views appropriate to their responsibilities. OWASP describes access control as mediating access to resources based on identity and policy (OWASP Foundation, n.d.). Role checks should be enforced by the server, not only by hiding interface elements.

Document upload and storage: Evidence upload is a core feature, but uploaded files create security and storage risks. OWASP recommends allowing only necessary file types, validating file types rather than trusting a supplied content-type value, limiting file size, generating safe stored filenames, and ensuring only authorized users can access uploads (OWASP Foundation, n.d.). TrackWIL should also use protected storage, controlled download links, appropriate backups and clear rules for retention and deletion.

Cloud hosting: Cloud hosting can make a web application accessible over the internet without the project team maintaining all underlying physical infrastructure. NIST describes cloud computing as on-demand network access to a shared pool of configurable computing resources (Mell and Grance, 2011). Hosting decisions should still consider cost, availability, backup and recovery, privacy, access controls and the location where student information is stored. Cloud hosting is a possible direction rather than a confirmed choice in Chapter 1.

Usability and testing: The application should make submission status easy to understand and minimise unnecessary steps for students and mentors. Testing should cover successful and unsuccessful logins, role permissions, document upload validation, submission and review status

changes, feedback, and progress reports. Evaluations should check whether users can complete these tasks reliably; claims of improved efficiency should be based on testing or user feedback rather than assumed in advance.

2.5 Research Gap

Existing placement-management systems and portfolio platforms already provide useful functions such as centralized placement information, paperwork, evidence sharing, assessment, communication and reporting. Therefore, the research gap is not that no digital WIL or placement tools exist.

The project-specific gap addressed by TrackWIL is the need for a focused, integrated workflow aligned with the requirements described in Chapter 1: students record daily WIL activities and upload evidence; mentors review submissions, provide feedback and approve or reject them; and coordinators and lecturers monitor verification status and identify outstanding work through role-specific dashboards.

Public product descriptions do not establish whether one existing platform, in the configuration available to the intended users, matches all these requirements in a simple workflow. The project will therefore design and test a solution against these requirements rather than claim that existing products cannot support them. The proposed contribution is a prototype tailored to the stated use case, with a shared record of submissions and their status.

The project evaluation should investigate whether users can submit and verify documentation more clearly and whether staff can identify outstanding work more easily. This gap and evaluation focus align with the aim and objectives in Chapter 1.

2.6 Conceptual / Technical Direction

TrackWIL will follow a centralized, role-based workflow. A student creates a dated WIL activity, completes the required documentation and attaches supporting evidence. The student submits the entry to the assigned workplace mentor. The mentor reviews it and either approves it or rejects it with feedback. The system records the current status so the student can see the outcome, while coordinators and lecturers use dashboards to identify submitted, pending, rejected, approved and outstanding work.

The main conceptual entities are User, Role, Student, Mentor Assignment, WIL Activity, Document/Evidence, Submission Review and Feedback. Relationships between these entities should be designed so each activity belongs to a student, relevant submissions can be reviewed by an authorized mentor, and status changes can be reported consistently. The design should avoid storing multiple conflicting copies of the same submission.

Technical priorities are usability, data integrity, access control, safe file handling, maintainability and testability. The group should choose the implementation framework, database and hosting service based on its actual development environment and skills; the source material currently available does

not confirm a specific stack. Regardless of the stack, the system should validate input, enforce permissions on the server, restrict file uploads and test each role's workflow.

2.7 Chapter Summary

This chapter reviewed research on electronic WIL and placement supervision, and compared InPlace, Sonia Online and portfolio/assessment tools such as Mahara and PebblePad. The review found useful patterns in centralized placement information, evidence collection, submission review, feedback, compliance and reporting. It also highlighted technical considerations involving relational data, authentication, role-based access, safe file uploads, hosting and usability.

The research gap is a project-specific integrated workflow that matches TrackWIL's stated needs for daily activity logging, evidence submission, mentor verification and role-specific progress monitoring. These findings inform the functional requirements and technical design in the next project stages. TrackWIL's effectiveness must be established through testing and evaluation rather than assumed.